Initial Data

Geometry Data

Height of structure	h	12	m
Depth of underground facing panel	ht	1,75	m
Horizontal distance between reinforcing elements	t	0,75	m
Verical distance between reinforcing elements	f	0,75	m

Soil properties

Unit weigt of backfill	!	20	kN/m^3
Internal friction angle of coarse backfill	"k	29	°
Cohesion of coase backfill	c'k	0	kPa
Friction angle between coarse backfill and steel reinforcing strips	μ	0,37	
cohesion of subsoil	c'k,A	80	kPa

Loads

Characteristic value of fill weight	g_k	30	kN/m^2
Characteristic value of vehicle load	q_k	24	kN/m^2

Other data

Steel quality		S275	kN/m^2
Corrosion coefficient of steel reinforcing strips	K_1	0,5	
	K_2	30	
	N	0.8	
Design lifetime	#	50	years

Critical failure surface parameters

With the given data of the internal friction (!k) of 38°, a linal interpolation is done between !k= 35° and !k=40° to get the value of the critical parameter value.

"k	"	D/h	K_c	K_q
25°	38°	0.43	0,26	0,70
30°	43.5°	0,37	0,21	0,60
29°	39.1°	0,382	0,22	0,62

Data for Critical failure surface

Earth pressure coefficient from max. normal force	K_c	0.22	
Earth pressure coefficient from surface	K_q	0.62	
	D = D/h*h	4.584	m
	$\varepsilon = 90° - (\omega - \varphi)$	21.9	°
Length of reinforcing strips	l = 0,8*h	9.6	m

Design method:

- Measure distance D from the top point of the structure. From the endpoint of this distance measure angle φ .
- Measure angle $ from the base point of the structure.
- The intersection of the two lines is point O.
- Distance R_a and R_b can be measured from the drawing.
- Points of the logarithmic spiral failure surface can be calculated by using the formular: $R_! = R_a * e^{\#!*\%\&'}$
- The " angle needs to be in radians while the φ must be in [°].
- The logarithmic spiral failure surface can be constructed by determining R_i values every 5° from 0° to ".

The results from this process are as follows:

ω	R_i
0	1384.5 cm
5	1452.5933 cm
10	1524.5862 cm
15	1600.1472 cm
20	1679.4531 cm
25	1762.6896 cm
30	1850.0514 cm
35	1941.7429 cm
39.1	2016.5 cm

Determination of earth pressures

In case of unloaded surface horizontal earth pressure (*h, B) increases linearly until depth H0, then it remains constant until the base level. In case of loaded surface horizontal earth pressure (*h,A) decreases linearly until H0 depth, then it remains constant until base level.

*h,A and *h, B stress need to be calculated at the level of every reinforcing element, then the covering curve (*h) need to be constructed.

Calculation oft he K_{cs} coefficient of earth pressure	K_{cs}	0.6194	
Depth associated with the maximum reinforcement force (limit depth)	H_0	4.2623	m
Value of lateral earth pressure at the surface	$\sigma h,A,F$	36.456	kN/m^2
	$\sigma h,B,F$	22.32	kN/m^2
Value of lateral earth pressure at the limit depth	$\sigma h,A,H_0$	18.7542	kN/m^2
	$\sigma h,A,H_0$	22.5051	kN/m^2

The horizontal earth pressure at each reinforcement level were measured from the earth pressure diagrams drawn in AutoCAD.

Analysis of sliding

The reinforced retaining wall should be considered as a single gravity wall.

$$\delta := \varphi \qquad \beta := 0 \qquad \alpha := 0$$

$$K_a := \frac{\cos(\delta)\cdot(\cos(\varphi+\alpha))^2}{(\cos(\alpha))^2\cdot\left(\sqrt{\cos(\alpha-\delta)}+\sqrt{\dfrac{\sin(\varphi+\delta)\cdot\sin(\varphi-\beta)}{\cos(\alpha+\beta)}}\right)^2} = 0.2692$$

Coefficient of active earth pressure	K_a	0.2692	
	β	0	°
	$\delta = \varphi$	29	°
	-	0	°
The characteristic value of earth pressures due to self weight	$P_{a,g,h,k}$	387.6854	kN/m
The characteristic value of earth pressures due to uniformly distributed surface load:	$P_{a,q,h,k}$	77.5371	kN/m
The design value of sliding force Ed	E_d	628.0503	kN/m

The design value of resistance

If sliding on cohesive soil surface	R_d	698.1818	kN/m
If sliding on granular soil surface	R_d	120.9402	kN/m
	G_k	240	kN/m

The smallest resistance is taken: $R_d = R_d =$ 120.9402 kN/m $< E_d =$ 628.0503 kN/m

The retaining wall is not sufficiently secured against sliding.

Analysis of Base Failure

Analysis of bearing capacity

The characteristic value of resistance	R_k	411.3274	kN/m
The design value of resistance	R_d	293.8053	kN/m
The design value of the load on the base of the retaining wall (effect of action)	V_d	396.9	kN/m

R_d = 293.8053 kN/m > V_d = 396.9 kN/m

The bearing capacity of the structure is not satisfactory.

The maximum height of the structure

The maximum height of the structure	h_{max}	8.1817	m

Horizontal earth pressure

level	level of reinforcing strips	Horizontal earth pressure	vertical stress	safty factor against pull out	sufficiency
		oh	ovi	k1	
	[m]	[kN/m2]	[kN/m2]		
1	0	36.456	30	0	sufficient
2	0,375	34.8708	37.5	1.42	sufficient
3	1,125	31.7004	52.5	1.64	sufficient
4	1,875	28.53	67.5	1.57	sufficient
5	2,625	25.3596	82.5	1.76	sufficient
6	3,375	22.499	97.5	1.81	sufficient
7	4,125	22.5051	112.5	2.15	sufficient
8	4,875	22.5051	127.5	1.69	sufficient
9	5,625	22.5051	142.5	1.97	sufficient
10	6,375	22.5051	157.5	2.28	sufficient
11	7,125	22.5051	172.5	2.63	sufficient
12	7,875	22.5051	187.5	3.01	sufficient
13	8,625	22.5051	202.5	3.44	sufficient
14	9,375	22.5051	217.5	3.92	sufficient
15	10,125	22.5051	232.5	4.45	sufficient
16	10.875	22.5051	247.5	5.04	sufficient
17	11.625	22.5051	262.5	5.7	sufficient
18	12	22.5051	277.5	6.06	Sufficient

Thickness of reinforcing strip

Level	Level of reinforcing strips	Thickness of reinforcimg strip	Increased width from corrosion	Calculated thickness of reinforcing strip	applied thickness of reinforcing strip
		vi	vk	vcpl,i	vapl,i
	[m]	[mm]	[mm]	[mm]	[mm]
1	0	0.5	5.4928	5.99	6
2	0,375	0.48	5.4928	5.97	6
3	1,125	0.58	5.4928	6.08	7
4	1,875	0.79	5.4928	6.28	7
5	2,625	0.88	5.4928	6.37	7
6	3,375	1.03	5.4928	6.53	7
7	4,125	1.04	5.4928	6.53	7
8	4,875	1.55	5.4928	7.05	8
9	5,625	1.55	5.4928	7.05	8
10	6,375	1.55	5.4928	7.05	8
11	7,125	1.55	5.4928	7.05	8
12	7,875	1.55	5.4928	7.05	8
13	8,625	1.55	5.4928	7.05	8
14	9,375	1.55	5.4928	7.05	8
15	10,125	1.55	5.4928	7.05	8
16	10,875	1.55	5.4928	7.05	8
17	11.625	1.55	5.4928	7.05	8
18	12	1.55	5.4928	7.05	8

Level	Level of reinforcing strip	Normal force(Charactaristic value)	Normal force design value	Distance	Cohesion length	Width of steel reinforcing strip	Applied width of steel reinforcing strip
		Nik	Ni,d	Di	D-Di	Si	Sapl
	(m)	(kN)	(kN)	(m)	(m)	(cm)	(cm)
T1	0	20.5065	27.6838	4.59	5.01	20	20
T2	0.375	19.6148	26.48	4.59	5.01	20	20
T3	1.125	17.8315	24.0725	4.59	5.01	13.6	15
T4	1.875	16.0481	21.665	4.54	5.06	9.4	10
T5	2.625	14.2648	19.2575	4.45	5.15	6.7	8
T6	3.375	12.6411	17.0654	4.32	5.28	4.9	6
T7	4.125	12.6557	17.0852	4.15	5.45	4.1	6
T8	4.875	12.6591	17.0898	3.94	5.66	3.5	4
T9	5.625	12.6591	17.0898	3.695	5.905	3	4
T10	6.375	12.6591	17.0898	3.41	6.19	2.6	4
T11	7.125	12.6591	17.0898	3.085	6.515	2.3	4
T12	7.875	12.6591	17.0898	2.725	6.875	2	4
T13	8.625	12.6591	17.0898	2.33	7.27	1.7	4
T14	9.375	12.6591	17.0898	1.89	7.71	1.5	4
T15	10.125	12.6591	17.0898	1.415	8.185	1.3	4
T16	10.875	12.6591	17.0898	0.89	8.71	1.2	4
T17	11.625	12.6591	17.0898	0.315	9.285	1	4
T18	12	12.6591	17.0898	0	9.6	0.98	4

$N_{pl.Rd}(x) := \frac{s_{apl.x} \cdot v_x \cdot f}{\gamma_{M0}}$ if $N_{i,d} \geq N_{pl,Rd}$

$N_{pl.Rd0} := \frac{0.2\ m \cdot 0.5\ mm \cdot f_y}{\gamma_{M0}} = 27.5\ kN$

$N_{pl.Rd1} := \frac{s_{apl.1} \cdot v_1 \cdot f_y}{\gamma_{M0}} = 26.48\ kN$

$N_{pl.Rd2} := \frac{s_{apl.2} \cdot v_2 \cdot f_y}{\gamma_{M0}} = 24.0725\ kN$

$N_{pl.Rd3} := \frac{s_{apl.3} \cdot v_3 \cdot f_y}{\gamma_{M0}} = 21.665\ kN$

$N_{pl.Rd4} := \frac{s_{apl.4} \cdot v_4 \cdot f_y}{\gamma_{M0}} = 19.2575\ kN$

$N_{pl.Rd5} := \frac{s_{apl.5} \cdot v_5 \cdot f_y}{\gamma_{M0}} = 17.0654\ kN$

$N_{pl.Rd6} := \frac{s_{apl.6} \cdot v_6 \cdot f_y}{\gamma_{M0}} = 17.0852\ kN$

$N_{pl.Rd7} := \frac{s_{apl.7} \cdot v_7 \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd8} := \frac{s_{apl.8} \cdot v_8 \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd9} := \frac{s_{apl.9} \cdot v_9 \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd10} := \frac{s_{apl.10} \cdot v_{10} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd11} := \frac{s_{apl.11} \cdot v_{11} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd12} := \frac{s_{apl.12} \cdot v_{12} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd13} := \frac{s_{apl.13} \cdot v_{13} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd14} := \frac{s_{apl.14} \cdot v_{14} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd15} := \frac{s_{apl.15} \cdot v_{15} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd16} := \frac{s_{apl.16} \cdot v_{16} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

$N_{pl.Rd0} := \frac{v_{17} \cdot s_{apl.17} \cdot f_y}{\gamma_{M0}} = 17.0898\ kN$

from here we can see that they're equal in every strip